PATRICK SLIZ

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EDUCATION

Harvard College, Cambridge, MA — GPA: 3.8/4.0 **May 2027**
A.B. Candidate in Computational Neuroscience. Secondary in Economics.

- Relevant Coursework: Computational Neuroscience, Biological Signal Processing, Memory Models, Linear Algebra

St. Sebastian's High School, Needham, MA — Valedictorian, 1560 SAT. **May 2023**

RESEARCH EXPERIENCE

Harvard Medical School / Massachusetts General Hospital, Department of Neurosurgery, Boston, MA **January 2026 - Present**
Undergraduate Researcher, Ziv Williams Laboratory

- Decode speech and language from intracranial single-neuron (Neuropixels) recordings in patients undergoing awake neurosurgical monitoring; build the analysis pipelines in Python
- Map transformer language-model representations onto neural firing to test how linguistic structure and meaning are encoded at single-cell resolution
- Diagnosed a generalization failure in the decoding pipeline as a word- vs. sentence-level representational mismatch; rebuilt the train/test framework to correct it

University of Warsaw, **Department of Quantitative Finance,** Warsaw, Poland **Summers 2024 - 2025**
Research Internship with Professor Piotr Wójcik

- Built ML models (XGBoost, LASSO) in R and Python predicting county- and state-level U.S. economic growth from night-time satellite imagery; applied explainable-AI methods to assess feature importance

Stanford University, Neurosciences Institute, HHMI Laboratory, Palo Alto, CA **Summers 2022 - 2023**
Software Developer in Cryo-electron Tomography (Cryo-ET) with Professor Axel Brunger

- Developed a subtomogram-averaging pipeline for high-resolution molecular imaging; translated MATLAB to Python and optimized 3D signal-processing routines with multi-threading and parallelization
- Built visualization, parsing, and cross-correlation tooling used by lab members to inspect reconstructions, working directly with structural biologists to turn experimental requirements into working software

PROFESSIONAL EXPERIENCE

GoTo, Boston, MA **June 2026 - August 2026**
Strategic Finance Intern, Business Planning & Analysis Team

- Built an ARR forecasting suite of six time-series and ML models (Prophet, ETS, MSTL, DHR-ARIMA, LightGBM, XGBoost) covering ~$100M in ARR; reduced forecast error (MAPE) from ~50% to ~14% on backtested holdout data and deployed to Databricks with an executive-facing UI and generative-AI insight summaries
- Shipped two production AI agents adopted internally: an accounting agent reconciling sales tax data across four sources into a single master file, replacing manual reconciliation; and a QBR agent generating deck cheat sheets and post-meeting analyses with human-in-the-loop gating
- Co-authored an AI strategy white paper with the VP of Strategic Finance on adoption priorities and AI pricing economics across GoTo's product portfolio; incorporated feedback from Product and Strategy stakeholders; reviewed by CFO and CCO

LEADERSHIP EXPERIENCE

The Harvard Independent, *Vice President* **January 2024 - Present**

- Co-lead operations and finances for a 140+ person organization alongside the President; own merchandise and program revenue
- Grew website readership >350% to 444k+ views in a single semester via SEO and infrastructure overhaul; built a Python pipeline automating weekly issue publication from Google Drive to WordPress

Harvard Undergraduate Venture Capital Group, *Sourcing Associate* **January 2024 - December 2025**

- Sourced 100+ startups and produced investment memos recommending companies to VC firms (Insight Partners, Republic)

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), R, MATLAB, C, Bash, Git
Methods: Time-series forecasting, Ridge/LASSO/logistic regression, PCA, SVM, LDA, permutation testing
Finance & Analytics: Excel, SQL, Power BI, Databricks, Stata, NetSuite
Languages: English (native), Polish (fluent), Spanish (conversational)